

# SHARP CUT-DELETION CLOCKS AND EXACT FIBRES FOR PREFIX-DIVISIBILITY DYNAMICS ON COMPOSITIONS

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**ABSTRACT.** For a positive composition  $a = (a_1, \dots, a_k)$  of  $N$ , retain the internal cut at prefix  $s_i = a_1 + \dots + a_i$  exactly when  $a_i$  divides  $s_i$ , delete all other cuts simultaneously, and iterate the resulting coarsening. We determine both its temporal structure and its complete one-step inverse problem. Every recurrent composition is fixed, and fixed points are counted by an explicit divisor-step recurrence. The maximum tail is zero for  $N \leq 3$  and  $N - 3$  for  $N \geq 4$ . Moreover,  $(1, 2, 1^{N-3})$  is the unique deepest composition, and its complete orbit is  $(1, 2 + t, 1^{N-3-t})$ . Independently, we encode a source as an increasing path through cut positions. For every labelled target, a no-skipped-target path recurrence gives its exact fibre; mandatory target cuts then factor this count into arithmetic interval factors. Positivity gives a complete image criterion, and the fibres sum to  $2^{N-1}$ . A paper-local exhaustive control checks all compositions through  $N = 18$ , making 3,408,240 exact assertions; this is regression pressure, not proof or novelty evidence. The direct-owner search remains bounded, so the manuscript is `HOLD_EXTERNAL`.

## 1. THE CUT FILTER AND ITS SUBTRACTION BOUNDARY

Let  $\mathcal{C}_N$  be the set of ordered tuples of positive integers with sum  $N$ . The familiar cut-set encoding identifies  $a = (a_1, \dots, a_k) \in \mathcal{C}_N$  with

$$(1) \quad \mathcal{D}(a) = \{s_1, \dots, s_{k-1}\} \subseteq [N-1], \quad s_i = \sum_{j=1}^i a_j.$$

Thus  $|\mathcal{C}_N| = 2^{N-1}$ , and deleting cuts is coarsening. We use this standard encoding and refinement language as background [2, 5].

Define  $F_N : \mathcal{C}_N \rightarrow \mathcal{C}_N$  by its retained cuts:

$$(2) \quad \mathcal{D}(F_N(a)) = \{s_i \in \mathcal{D}(a) : a_i \mid s_i\}.$$

All tests in (2) use the parts of the composition at the start of the epoch, and all failing cuts are deleted simultaneously. The final endpoint  $N$  is not a cut and is never tested. For example,

$$(3) \quad (1, 2, 1, 1, 1) \mapsto (1, 3, 1, 1) \mapsto (1, 4, 1) \mapsto (1, 5).$$

Restricted compositions and local conditions are established enumerative themes [1, 3]. A nearby static divisibility class asks whether the *index*  $i$  divides the prefix sum [4]; our dynamic test instead asks whether the current *part*  $a_i$  divides its endpoint and then changes the parts by coarsening.

We assign the composition–subset bijection, refinement order, elementary divisibility, and generic path dynamic programming zero contribution credit. Internally, the mechanism is not the balanced part splitting of P126, the equal-run merging of P147, the Euclidean quotient rotation of P131, the set-partition token transfer of P169, the permutation prefix reversal of P181, the prefix-diversity word map of P185, or the support-rank compression of P186. This subtraction is not an ownership result. The external search was bounded; a non-hit is not evidence of novelty, priority, completeness, or freedom to operate.

## 2. FIXED STATES AND THE SHARP CLOCK

Write  $\mu(a)$  for the least  $t \geq 0$  such that  $F_N^t(a)$  is recurrent.

**Proposition 2.1** (recurrence and fixed-state census). *Every recurrent state is fixed, and*

$$(4) \quad a \in \text{Fix}(F_N) \iff a_i \mid s_i \quad (1 \leq i < k).$$

Set  $A(0) = 1$  and, for  $v \geq 1$ , set

$$(5) \quad A(v) = \sum_{\substack{0 \leq u < v \\ v-u \mid v}} A(u).$$

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Then, for every  $N \geq 1$ ,

$$(6) \quad |\text{Fix}(F_N)| = \sum_{v=0}^{N-1} A(v).$$

*Proof.* Equation (2) gives  $\mathcal{D}(F_N(a)) \subseteq \mathcal{D}(a)$ , with equality precisely under (4). Every nonfixed epoch strictly loses a cut, so a cycle cannot contain a nonfixed state.

For the count, regard a sequence of internal cuts as an increasing path from 0. If the last cut is  $v$  and its predecessor is  $u$ , the part ending at  $v$  has size  $v - u$ ; fixedness at that cut is exactly  $v - u \mid v$ . Last-edge decomposition gives (5). A fixed composition of  $N$  has either no internal cut, represented by  $v = 0$ , or a unique last internal cut  $v < N$ . The final part  $N - v$  is untested, which proves (6).  $\square$

**Theorem 2.2** (sharp transient and unique extremizer). *For  $N \leq 3$ , every composition is fixed. For every  $N \geq 4$ ,*

$$(7) \quad \max_{a \in \mathcal{C}_N} \mu(a) = N - 3,$$

*and the unique composition attaining this maximum is*

$$(8) \quad \omega_N = (1, 2, 1^{N-3}).$$

*At every time  $0 \leq t \leq N - 3$  its state is*

$$(9) \quad F_N^t(\omega_N) = (1, 2 + t, 1^{N-3-t}).$$

*Proof.* If a composition has an internal first cut, that cut always survives: its ending part and endpoint are both  $a_1$ . A composition of length  $k$  therefore has at most  $k - 2$  deletable cuts. Every nonfixed epoch deletes at least one. The only composition of length  $N$  is  $(1^N)$ , which is fixed, so a nonfixed state has  $k \leq N - 1$  and hence tail at most  $N - 3$ . Directly from the rule, all compositions for  $N = 1, 2, 3$  are fixed.

For  $\omega_N$ , suppose (9) holds at time  $t < N - 3$ . The middle part  $2 + t$  ends at  $3 + t$ . It does not divide this endpoint, because otherwise it would divide their difference 1. The first part and all trailing unit parts pass their tests, so exactly the cut after the middle part is deleted. This proves (9) by induction and shows that  $\mu(\omega_N) = N - 3$ .

It remains to prove uniqueness. Equality in the two upper bounds forces an extremizer to have  $N - 1$  parts and to lose exactly one cut at every nonfixed epoch. A positive composition of  $N$  with  $N - 1$  parts has one part equal to 2 and all others equal to 1. It is fixed if the 2 is first or final. Otherwise let  $r \geq 1$  be the number of leading unit parts. Throughout the orbit, cuts ending unit parts survive; only the cut after the block containing the 2 can disappear. There are just  $N - r - 2$  parts to that block's right, so its tail is at most  $N - r - 2$ . Equality with  $N - 3$  forces  $r = 1$ , giving exactly (8).  $\square$

The fixed counts for  $N = 1, \dots, 15$  are

$$1, 2, 4, 7, 13, 20, 37, 55, 97, 157, 267, 378, 719, 1061, 1763.$$

These values follow from (5); they are quoted only as small controls, not as a replacement for the recurrence.

### 3. EVERY-TARGET FIBRES

Fix a target  $b \in \mathcal{C}_N$  with cut set  $T = \mathcal{D}(b)$ . A source composition is uniquely an increasing path

$$(10) \quad 0 = x_0 < x_1 < \dots < x_\ell = N,$$

whose nonfinal vertices are its cuts. Declare an edge  $u \rightarrow v$  to be  $T$ -admissible when

- (i) no member of  $T$  lies strictly between  $u$  and  $v$ ; and
- (ii) if  $v < N$ , then  $v \in T$  holds exactly when  $v - u \mid v$ .

No divisibility condition is imposed when  $v = N$ .

**Theorem 3.1** (global target-path recurrence). *Let  $P_T(0) = 1$  and, for  $1 \leq v \leq N$ , define*

$$(11) \quad P_T(v) = \sum_{\substack{0 \leq u < v \\ u \rightarrow v \text{ } T\text{-admissible}}} P_T(u).$$

*Then every labelled target has the exact fibre size*

$$(12) \quad |F_N^{-1}(b)| = P_T(N).$$

*Consequently  $b \in \text{im}(F_N)$  if and only if  $P_T(N) > 0$ .*

*Proof.* Every target cut must already occur on a source path, so a source edge cannot jump over it; this is condition (i). At a nonfinal source vertex  $v$ , the incoming part is  $v - u$ . Equation (2) retains  $v$  precisely when this step divides  $v$ . Thus condition (ii) says simultaneously that each target vertex survives and that each extra source vertex is deleted. The final endpoint is untested. It follows that source paths mapping to  $b$  are exactly the  $T$ -admissible paths. Last-edge decomposition gives (11), proving the fibre and image statements.  $\square$

The mandatory target cuts make this inverse count local. List all target endpoints as

$$0 = t_0 < t_1 < \dots < t_m = N,$$

so  $T = \{t_1, \dots, t_{m-1}\}$ . For consecutive endpoints  $p < q$ , put  $h_p(p) = 1$  and, for  $p < v < q$ , put

$$(13) \quad h_p(v) = \sum_{\substack{p \leq u < v \\ v-u \nmid v}} h_p(u).$$

Define an internal and a final interval factor by

$$(14) \quad K(p, q) = \sum_{\substack{p \leq u < q \\ q-u \mid q}} h_p(u) \quad (q < N),$$

$$(15) \quad K_*(p, N) = \sum_{p \leq u < N} h_p(u).$$

**Corollary 3.2** (factorized fibre and complete image). *For every target endpoint list as above,*

$$(16) \quad |F_N^{-1}(b)| = \left( \prod_{j=1}^{m-1} K(t_{j-1}, t_j) \right) K_*(t_{m-1}, N).$$

*In particular, the target is in the image exactly when every factor in (16) is positive. The empty product covers the one-part target.*

*Proof.* Within an interval  $(p, q)$ , every extra source vertex must be deleted. Recurrence (13) counts exactly the paths from  $p$  to  $v$  whose intervening vertices, including  $v$ , have nondividing incoming steps. For an internal target endpoint  $q$ , the last step must instead divide  $q$ , giving (14); at  $N$  no test is made, giving (15). Restriction to consecutive target intervals and concatenation are inverse bijections. Multiplying their independent path counts proves (16); positivity is therefore equivalent to image membership.  $\square$

**Corollary 3.3** (fibre mass). *For every  $N \geq 1$ ,*

$$(17) \quad \sum_{b \in \mathcal{C}_N} |F_N^{-1}(b)| = 2^{N-1}.$$

*Proof.* The labelled fibres partition the source carrier  $\mathcal{C}_N$ , and the cut-set encoding gives  $|\mathcal{C}_N| = 2^{N-1}$ .  $\square$

Both (11) and the complete collection of interval recurrences can be evaluated with  $O(N^2)$  arithmetic operations for a fixed target. The global recurrence exposes the inverse bijection directly, while the product formula separates the contribution of each mandatory target cut.

#### 4. EXACT PRESSURE, LIMITATIONS, AND CONCLUSION

The paper-local standard-library verifier exhausts all  $2^{N-1}$  compositions for every  $1 \leq N \leq 18$ . It constructs the literal map, decomposes its functional graph, tests the fixed recurrence and every time slice of the extremal orbit, and compares true indegrees target by target with both (11) and (16). It checks image equality and fibre mass only after the pointwise comparisons. Selected complete boxes appear in Table 1.

| $N$ | states | image | fixed | max tail | deepest | max fibre |
|-----|--------|-------|-------|----------|---------|-----------|
| 4   | 8      | 7     | 7     | 1        | 1       | 2         |
| 8   | 128    | 73    | 55    | 5        | 1       | 11        |
| 12  | 2048   | 801   | 378   | 9        | 1       | 59        |
| 15  | 16384  | 4906  | 1763  | 12       | 1       | 182       |
| 18  | 131072 | 28535 | 7398  | 15       | 1       | 696       |

TABLE 1. Selected exhaustive author controls. “Deepest” counts states of maximum tail. Proofs, rather than the finite range, establish the theorems.

The results concern positive compositions, simultaneous updates, and the literal divisibility predicate in (2). We do not give a closed scalar formula for the full image size, arbitrary pointwise tails, or time- $t$  fibres. We also make no theorem for weak compositions, asynchronous deletions, other divisibility directions, or weighted refinements. The finite computation is not proof, experiment, or ownership evidence. What is settled here is the all-parameter fixed and sharp-clock structure together with a complete, computable one-step inverse atlas. External circulation remains blocked while ownership is amber.

#### DECLARATIONS

Data availability. No external data were used. Exact source code and its canonical stdout accompany the manuscript.

Ethics. The work studies a finite combinatorial map and raises no human-subject, personal-data, or deployment issue.

CRedit roles. Anonymous author(s) performed conceptualization, formal analysis, software, validation, original drafting, and review and editing.

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